Inference and interpretation of long-term changes in infectious disease transmission using semi-mechanistic models

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Abstract

Introduction

A fundamental aim in infectious disease dynamics research is characterizing how disease transmission rates change over time. For example, classical studies of endemic childhood infections, particularly measles, focused on estimating seasonal fluctuation in transmission rates driven by school term and climate effects. These efforts not only generated key insights into the drivers of epidemic dynamics but also established core methodological foundations [1, 2, 3, 4]. Yet transmission rates can also evolve over longer time scales, reflecting gradual shifts in host behavior, demography, socioeconomic factors, and environmental conditions. Compared with seasonal variation, however, these low-frequency changes have received far less attention, in part because they are more difficult to identify from incidence data and are often confounded with other slow-moving processes such as susceptible replenishment and vaccination [5].

Disruptions in disease transmission caused by COVID-19 nonpharmaceutical interventions also highlighted the similar need for new frameworks for estimating transmission rates that can vary flexibly across two different time scales: explaining these disruptions requires simultaneous estimation of seasonal transmission patterns and rapid changes in transmission rate reflecting intervention effects. To capture the impact of nonpharmaceutical interventions on disease transmission, some researchers have relied on known metrics of intervention strengths, such as mobility patterns [6, 7] and stringency indices [8], whereas other have sought to directly estimate these impacts through various statistical methods [9, 10, 11]. Notably, models utilizing Bayesian inference have proven particularly useful for estimating flexible changes in transmission patterns, where statistical constraints through prior distributions can appropriately limit the variability in parameter estimates [12, 13, 14]. These methodological advances provide a basis for developing semi-mechanistic models that allow estimation of transmission rates across two different time scales, with potential applications beyond the context of COVID-19 interventions, but their estimability and generalizability have not yet been tested. Furthermore, the apparent discrepancy between the estimated intervention effects and observed changes in mobility patterns indicate potential challenges in interpreting parameter estimates from semi-mechanistic models [12].

Here, we present a semi-mechanistic framework for simultaneously estimating seasonal and long-term variation in disease transmission and investigate its estimability and interpretibility. We test the resulting framework against a 126-year long time series of smallpox deaths, which presents a computationally challenging example.

Materials and Methods

Data

We used a weekly time series of smallpox deaths to test the ability of a semi-mechanistic model to infer long-term changes in disease transmission (Figure 1). The original data set spans from 1664 to 1930 and was digitized from London Bills of Mortality and the Registrar General's Weekly Returns by [15]. Here, we considered mortality records between 1670 and 1795 to focus on the dynamics before the discovery of vaccination. Even in the absence of vaccination, an increase in variolation to protect against severe outcomes (Figure 1A) and a rapid increase in population sizes due to increasing births (Figure 1B,C) would have caused complex changes in transmission conditions over the 126-year period, which cannot be explained by traditional modeling approaches that only account for seasonal transmission forcing. Data on population sizes and birth rates were obtained from the publicly available data set from [15]. Death rates were calculated to match increasing rates of population sizes: specifically, we calculated the difference between the weekly rate of population size increase and weekly birth rate to obtain weekly death rates.

Mathematical model

To focus on the feasibility of estimating long-term changes in transmission, we decided to use the simplest transmission model, namely the SIR model. Assuming density dependent transmission, the discrete time SIR model can be written as:

$$\mathrm{FOI}(t) = \beta(t)(I(t) + \omega) \tag{1}$$

$$\Delta S(t) = \left[1 - \exp(-(\mathrm{FOI}(t) + d(t))\Delta t)\right] S(t - \Delta t) \tag{2}$$

$$N_{SI}(t) = \frac{\mathrm{FOI}(t)\Delta S(t)}{\mathrm{FOI}(t) + d(t)} \tag{3}$$

$$\Delta I(t) = \left[1 - \exp(-(\gamma + d(t))\Delta t)\right] I(t - \Delta t) \tag{4}$$

$$N_{IR}(t) = \frac{\gamma \Delta I(t)}{\gamma + d(t)} \tag{5}$$

$$S(t) = S(t - \Delta t) + b(t)N - \Delta S(t) \tag{6}$$

$$I(t) = I(t - \Delta t) - \Delta I(t) + N_{SI}(t) \tag{7}$$

where $\mathrm{FOI}(t)$ represents the force of infection; $\beta(t)$ represents the transmission rate; ω represents the number of imported infections; γ represents the recovery rate; $d(t)$ represents the death rate; and $b(t)$ represents the birth rate. The transmission rate is decomposed into a seasonal term $\beta_{\mathrm{seas}}(t)$ and a multiplicative time-varying transmission term $\delta(t)$:

$$\beta(t) = \frac{\beta_{\mathrm{seas}}(t)\delta(t)}{N_{1670}}, \tag{8}$$

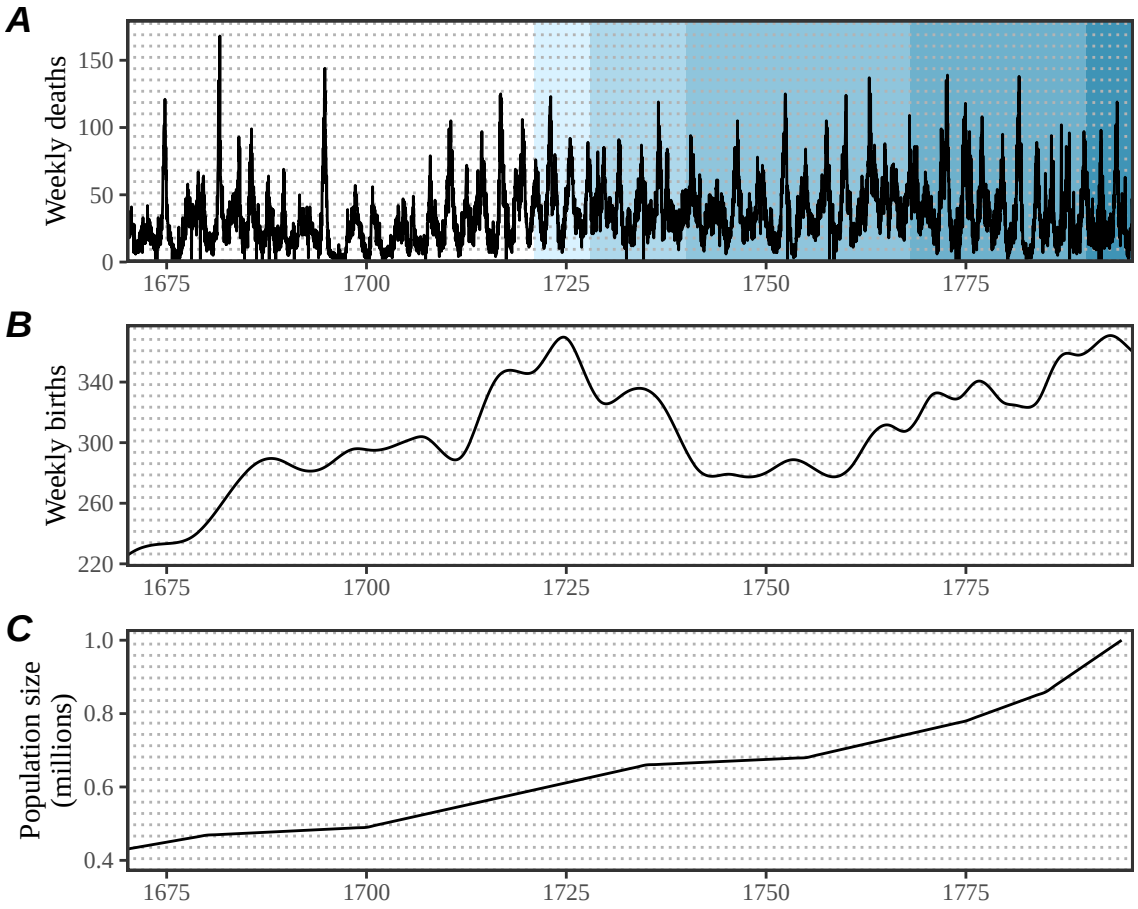


Figure 1: **Data**

where N_{1670} represents the population size at the beginning of 1670, which is included to scale the $\beta_{\text{seas}}(t)$. Previous studies have illustrated the feasibility of estimating $\delta(t)$ at different time scales by anchoring the seasonal term $\beta_{\text{seas}}(t)$ to certain periods. For example, when inferring the impact of COVID-19 interventions, $\delta(t)$ was set to 1 during the pre-pandemic period and allowed to vary from 2020 onward, with change points defined at intervals ranging from one to four weeks [12, 13, 14]. Since we are interested in capturing long-term changes in transmission, we allowed $\delta(t)$ to vary every year.

Statistical inference and implementation

In order to simultaneously estimate $\beta_{\text{seas}}(t)$ and $\delta(t)$ alongside other parameters and initial conditions, we needed to appropriately constrain the parameter space. Otherwise, estimating 52 terms for $\beta_{\text{seas}}(t)$ and 126 terms for $\delta(t)$ would require estimation of > 170 parameters, which is more than an order of magnitude greater than the number of parameters that are estimated for a typical infecitous disease model.

Bayesian inference, implemented via the Stan software, enables us to constrain the parameter space through prior distributions and allows for fast, efficient inference through state-of-art inferece algorithms [16]. We first discuss our model parameterization, prior choices, and observation models used for fitting. Then, we describe our implementation in Stan.

If we were to estimate all 52 terms of the seasonal term $\beta_{\text{seas}}(t)$ separately, then the resulting transmission rate may not be realistic and difficult to interpret. Thus, to constrain the shape of the seasonal term $\beta_{\text{seas}}(t)$ and allow for biological interpretation, we imposed a cyclic, random-walk prior on a log scale [14]:

$$\log \beta_{\text{seas}}(t+1) \sim \text{Normal}(\log \beta_{\text{seas}}(t), \sigma_{\text{seas}}), \quad t = 1, \ldots, 51 \tag{9}$$

$$\log \beta_{\text{seas}}(1) \sim \text{Normal}(\log \beta_{\text{seas}}(52), \sigma_{\text{seas}}) \tag{10}$$

where the standard deviation σ_{seas} determines the smoothness of the seasonal transmission rate. Strong priors were imposed to constrain the parameter space for σ_{seas}:

$$\sigma_{\text{seas}} \sim \text{Half-Normal}(0, 0.01). \tag{11}$$

We further imposed an informative prior on $\beta_{\text{seas}}(t)$, reflecting previously reported $\mathcal{R}_0$ values for smallpox transmission:

$$\log \beta_{\text{seas}}(t) \sim \text{Normal}(0.5, 0.1). \tag{12}$$

Combined with the assumed recovery rate of $-\log(1-1/3)$/weeks, which corresponds to a mean infectious period of 3 weeks, the prior distribution of the basic reproduction number has a median of 4.9 and a 95% quantile of 4.1 to 6.0.

For similar reasons as above, we also utilized random walk priors to model the multiplicative time-varying transmission term on a log scale: $\delta(t) = \exp(\epsilon(t))$. In order to reduce correlations in posterior distributions, we parameterized $\epsilon(t)$ by expression it as a sum of independent normal variables $z_\epsilon(t)$:

$$z_\epsilon(t) \sim \text{Normal}(0, 1), \quad t = 1671, \ldots, 1795 \tag{13}$$

$$\sigma_\epsilon \sim \text{Half-Normal}(0, 0.01) \tag{14}$$

$$\epsilon(1670) = 0 \tag{15}$$

$$\epsilon(t) = \epsilon(t-1) + \sigma_\epsilon z_\epsilon, \quad t = 1671, \ldots, 1795 \tag{16}$$

$$\delta(t) = \exp(\epsilon(t)) \tag{17}$$

where $z_\epsilon(t)$ determines the smoothness of $\epsilon(t)$. By setting the initial $\epsilon(1670)$ to zero and imposing constraining priors on σ_ϵ, this approach anchors the seasonal transmission term $\beta_{\text{seas}}(t)$ to early periods and allow for slow variation in transmission rates over time.

The model was fitted to smallpox deaths $D(t)$ by assuming a negative binomial observation model:

$$D(t) \sim \text{NegBin}(\rho N_{IR}(t), \phi) \tag{18}$$

$$\rho \sim \text{Beta}(1, 1) \tag{19}$$

$$\phi \sim \text{Normal}(100, 20) \tag{20}$$

where ρ represents the infection mortality rate, and ϕ represents the overdispersion parameter. Alongside parameters described earlier, we also estimated the initial conditions $S(1670)$ and $I(1670)$ at the beginning of 1670 as well as imported infections ω:

$$S(1670) \sim \text{Beta}(1, 1) \tag{21}$$
$$I(1670) \sim \text{Half-Normal}(0, 0.001) \tag{22}$$
$$\omega \sim \text{Half-Normal}(0, 200) \tag{23}$$

Previous studies primarily relied on Hamiltonian Monte Carlo (HMC) sampling implemented in Stan to estimate parameters of transmission models [12, 13, 14]. While this approach enables efficient exploration of high-dimensional parameter spaces and provides accurate uncertainty quantification, inference can remain computationally demanding for nonlinear models with many parameters. During our investigation, we also encountered convergence issues, partly reflecting the sensitivity of HMC sampling to initial conditions. Instead, we began with an optimization from 1000 random starting values to obtain the maximum a posteriori estimates. Then, we used Laplace approximation to obtain approximate posterior distributions.

Simulation studies

Baseline simulation. To evaluate the ability of a semi-mechanistic model to correctly infer seasonal and long-term variation in transmission rates, we generated a synthetic data set using median posterior estimates from the smallpox analysis.

Variolation simulation.

Results

Discussion

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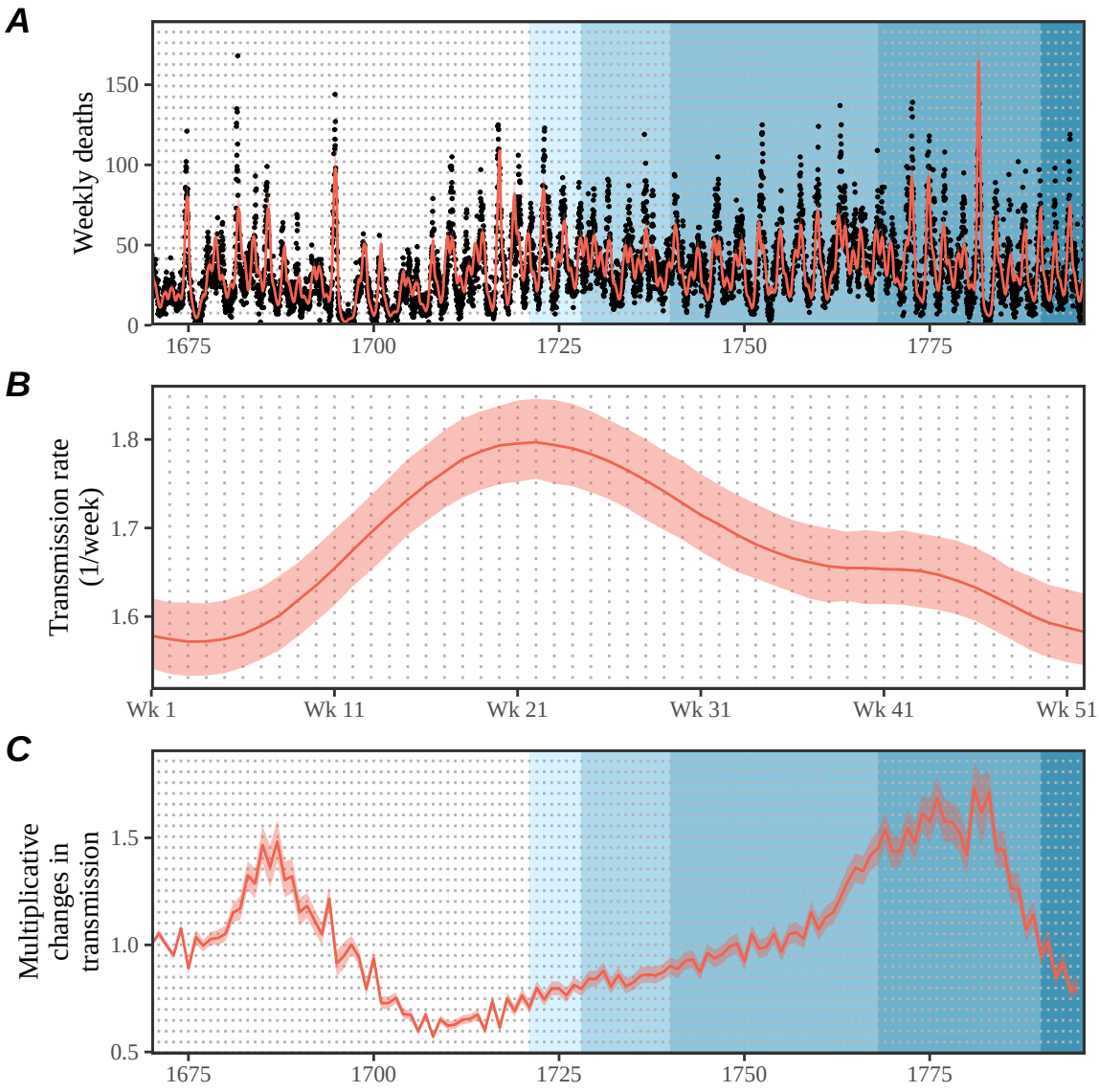


Figure 2: **Data**

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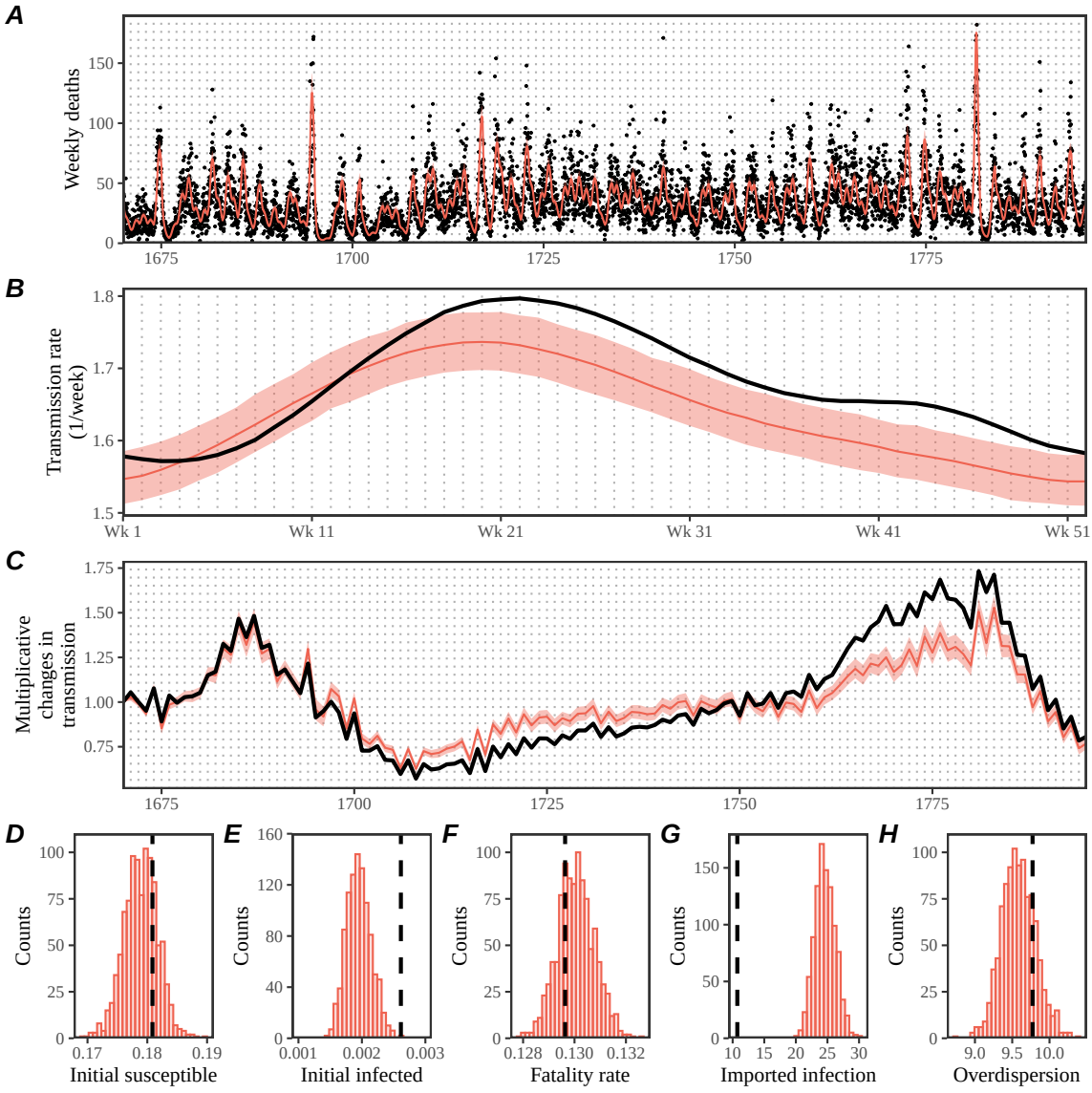


Figure 3: **Data**

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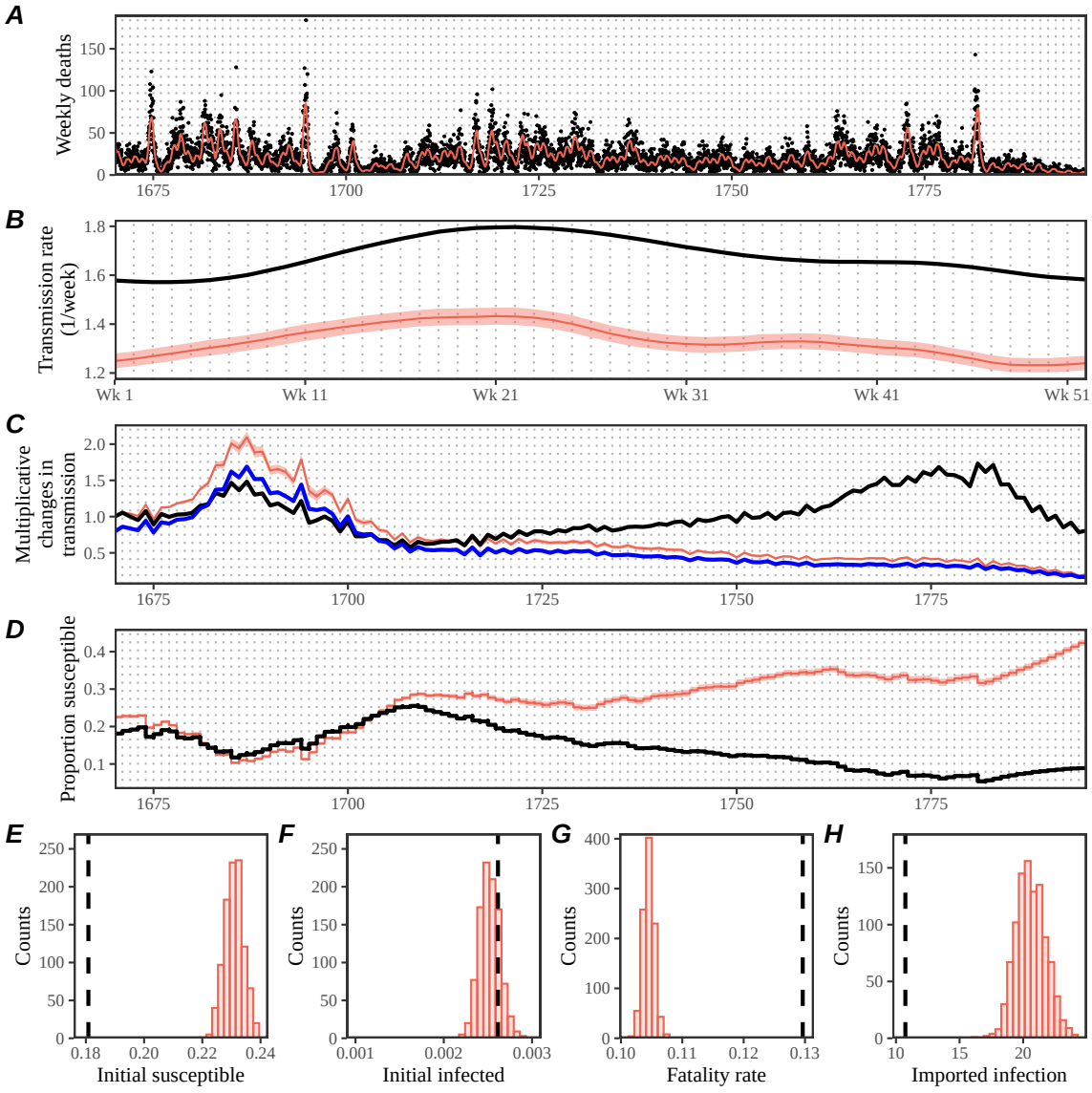


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